

Telco Customer Churn Analysis

Summary And Recommendations

Project Overview

This project analyzes customer churn behavior in a telecom company using the Telco Customer Churn dataset. The goal of the analysis was to identify the major factors contributing to customer churn and generate business insights that can help improve customer retention strategies.

The project involved:

- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Percentage-Based Analysis
- Statistical Visualization
- Customer Behavior Analysis

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used extensively for analysis and visualization.

Dataset Overview

The dataset consists of customer demographic information, telecom services, billing details, contract information, and churn status.

Total Customers:

- **7043 customers**

Churn Distribution:

Customer Status	Count	Percentage
Retained Customers	5174	73.5%

Churned Customers	1869	26.5%
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Key Insight:

Approximately **1 in every 4 customers left the telecom company**, indicating a considerable customer retention problem.

The churn distribution pie chart clearly highlighted that while the majority of customers remained loyal, the churn percentage was still significantly high.

Data Cleaning & Preparation

Several preprocessing techniques were applied before analysis:

Steps Performed:

- Checked missing values
- Converted `TotalCharges` datatype from object to numeric
- Removed blank records
- Verified duplicate records
- Standardized categorical variables for readability

Example:

`SeniorCitizen`

- `0` → No
- `1` → Yes

These preprocessing steps improved dataset consistency and analytical accuracy.

Demographic Analysis

Gender-Based Churn Analysis

The gender analysis showed that churn behavior was nearly identical between male and female customers.

Findings:

- Male Customers: approximately **50%**
- Female Customers: approximately **50%**
- Churn distribution remained nearly equal across genders

Conclusion:

Gender does not significantly influence customer churn behavior.

Senior Citizen Analysis

Senior citizens showed a noticeably higher churn percentage compared to non-senior customers.

Percentage Findings from Charts:

- Senior citizens had approximately **41–42% churn**
- Non-senior customers had approximately **23–24% churn**

Key Insight:

Senior citizens were almost **twice as likely to churn** compared to younger customers.

Possible Reasons:

- Difficulty adapting to telecom services
- Lack of technical support
- Higher pricing sensitivity

Recommendation:

The company should:

- Improve support accessibility
- Offer simplified plans
- Create senior-focused customer assistance programs

The stacked percentage bar charts clearly highlighted this large difference in churn behavior.

Tenure Analysis

Customer tenure emerged as one of the strongest indicators of churn.

Key Findings:

- Customers with tenure below 12 months showed the highest churn probability
- Customers with long tenure demonstrated strong loyalty behavior

Percentage Insight:

- Most churned customers belonged to the early subscription period
- Long-term customers showed significantly lower churn percentages

Business Interpretation:

Customer retention is most critical during the first few months after subscription.

Recommendation:

The company should:

- Strengthen onboarding programs
- Increase customer engagement in the first 90 days
- Offer introductory discounts and loyalty benefits

The histogram analysis clearly showed that churn decreases as customer tenure increases.

Contract Type Analysis

Contract type demonstrated one of the strongest relationships with customer churn.

Month-to-Month Contracts

Percentage Findings:

- Approximately **42–43%** of month-to-month customers churned

Insight:

Customers without long-term commitments were far more likely to switch providers.

One-Year Contracts

Percentage Findings:

- Churn rate was approximately **11–12%**

Insight:

One-year contracts significantly improved customer retention compared to month-to-month plans.

Two-Year Contracts

Percentage Findings:

- Churn rate was only around **2–3%**

Key Insight:

Two-year contract customers were the most loyal customer segment.

Business Recommendation:

The company should encourage customers to migrate toward:

- Annual contracts
- Two-year plans
- Loyalty subscriptions

The contract-type stacked percentage charts clearly showed the dramatic reduction in churn with longer contracts.

Service-Based Analysis

Multiple telecom services were analyzed using subplot countplots and percentage-based stacked bar charts.

Internet Service Analysis

Fiber Optic Customers

Percentage Findings:

- Fiber optic users showed approximately **41–42% churn**

Insight:

Fiber optic customers represented one of the highest churn groups despite using premium internet services.

Possible Reasons:

- High pricing
 - Service dissatisfaction
 - High customer expectations
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DSL Customers

Percentage Findings:

- DSL users showed approximately **19% churn**

Insight:

DSL customers demonstrated better retention compared to fiber optic users.

Online Security Analysis

Online Security was one of the strongest customer retention indicators.

Percentage Findings:

Online Security Status	Approximate Churn
No Online Security	~42%

Uses Online Security ~15%

Key Insight:

Customers without Online Security were almost **3 times more likely to churn**.

Business Impact:

Value-added security services significantly improve customer retention.

Tech Support Analysis

Tech Support showed a major influence on customer loyalty.

Percentage Findings:

Tech Support Status	Approximate Churn
No Tech Support	~42%
Uses Tech Support	~15%

Insight:

Customers receiving technical support were much more likely to stay with the company.

Recommendation:

The company should:

- Improve support quality
 - Promote support services
 - Offer affordable support plans
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Online Backup & Device Protection

Customers using:

- Online Backup
- Device Protection

showed lower churn percentages compared to customers without these services.

Percentage Insight:

Customers without these services consistently showed churn rates above **35–40%**, while customers using these services showed much lower churn percentages.

Interpretation:

Customers using multiple services become more integrated into the telecom ecosystem and are less likely to leave.

Streaming Services Analysis

StreamingTV and StreamingMovies services showed moderate relationships with churn.

Findings:

- Customers without streaming services showed higher churn
- Streaming service users demonstrated comparatively better retention

Interpretation:

Streaming services positively contribute to customer engagement but are not the strongest churn indicators individually.

Payment Method Analysis

Payment method was another important churn factor.

Electronic Check Users

Percentage Findings:

- Electronic Check customers showed approximately **45% churn**

Key Insight:

This was one of the highest churn percentages observed in the analysis.

Possible Reasons:

- Manual payment inconvenience
 - Failed payment issues
 - Lower customer engagement
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Automatic Payment Users

Customers using:

- Credit cards
- Bank transfers
- Auto-pay methods

showed significantly lower churn percentages.

Percentage Insight:

Automatic payment users showed churn percentages below **20%** in most cases.

Recommendation:

The company should encourage automatic billing adoption through:

- Discounts
 - Cashback offers
 - Loyalty incentives
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Monthly Charges Analysis

Findings:

- Customers with higher monthly charges were more likely to churn
- Customers paying lower monthly charges showed stronger retention

Percentage Insight:

High-paying customers formed a significant portion of churned users, especially among month-to-month subscribers.

Interpretation:

Customers are willing to pay higher prices only when service quality and satisfaction remain high.

Visualization Summary

The project included several impactful visualizations:

Visualizations Created:

- Countplots
- Pie Charts
- Histograms
- Stacked Percentage Bar Charts
- Multi-Subplot Service Comparison Charts

Visualization Benefits:

- Simplified customer segmentation
- Easy churn comparison across services
- Better understanding of high-risk customers
- Improved business storytelling

The percentage-based stacked bar charts were particularly effective in visualizing churn proportions across customer groups.

Major High-Risk Customer Groups Identified

Customers with the following characteristics showed the highest churn percentages:

Customer Category	Approximate Churn
Month-to-Month Contracts	~43%
Electronic Check Users	~45%
No Online Security	~42%
No Tech Support	~42%

Fiber Optic Users	~41%
Senior Citizens	~42%

Business Recommendations

1. Improve Early Customer Retention

Since most churn occurs during the initial months:

- Improve onboarding
 - Increase engagement
 - Offer welcome incentives
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2. Promote Long-Term Contracts

Encourage migration toward:

- One-year contracts
- Two-year subscriptions

through loyalty rewards and discounted pricing.

3. Increase Adoption of Value-Added Services

Promote:

- Online Security
- Tech Support
- Device Protection
- Online Backup

These services demonstrated significantly lower churn percentages.

4. Improve Senior Citizen Support

Provide:

- Simplified plans
 - Personalized assistance
 - Dedicated support channels
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5. Encourage Auto-Payment Systems

Reduce dependency on Electronic Checks and promote:

- Digital payments
 - Recurring billing systems
 - Automated payment incentives
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Conclusion

This Telco Customer Churn Analysis successfully identified the primary drivers influencing customer churn using detailed exploratory data analysis and percentage-based visualizations.

Final Key Insights:

- Overall churn rate: **26.5%**
- Month-to-month contract churn: **~43%**
- Electronic check churn: **~45%**
- Senior citizen churn: **~42%**
- Customers without Online Security or Tech Support churned at rates above **40%**
- Two-year contract customers showed the strongest retention with churn below **3%**

The project effectively combined data cleaning, exploratory analysis, percentage visualization, and business interpretation to generate actionable insights. These findings can help telecom companies reduce churn, improve customer satisfaction, and build more effective customer retention strategies.